

Exercise 4.1:

Consider minimizing the cost function $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T\mathbf{A}\mathbf{x} - \mathbf{b}^T\mathbf{x}$ with respect to the design variable $\mathbf{x}$, where $\mathbf{A}$ is a positive definite matrix.

Denote the initial solution guess as $\mathbf{x}_0$. Assume that at $\mathbf{x}_0$, an exact line search is carried out along the search direction $\mathbf{p}_0 = -\nabla f(\mathbf{x}_0)$.

Compute the step size a_0 .

Solution:

It is easy to show that $\mathbf{p}_0 = -\nabla f(\mathbf{x}_0) = -(\mathbf{A}\mathbf{x}_0 - \mathbf{b})$.

The exact line search finds the step size a_0 by minimizing $f(\mathbf{x}_0 + a_0\mathbf{p}_0)$ with respect to a_0 . The loss function is thus

$$\begin{aligned} f(\mathbf{x}_0 + a_0\mathbf{p}_0) &= \frac{1}{2}(\mathbf{x}_0 + a_0\mathbf{p}_0)^T\mathbf{A}(\mathbf{x}_0 + a_0\mathbf{p}_0) - \mathbf{b}^T(\mathbf{x}_0 + a_0\mathbf{p}_0) \\ &\propto \frac{1}{2}a_0^2\mathbf{p}_0^T\mathbf{A}\mathbf{p}_0 + a_0\mathbf{p}_0^T\mathbf{A}\mathbf{x}_0 - a_0\mathbf{b}^T\mathbf{p}_0, \end{aligned} \quad (1)$$

where all the terms not depending on a_0 have been ignored. Note that since $\mathbf{A}$ is positive definite, $\mathbf{p}_0^T\mathbf{A}\mathbf{p}_0$ is positive. Finding the minimizer for (1) can be achieved by taking the derivative with respect to a_0 and setting the result to zero, which yields

$$a_0 = -\frac{\mathbf{p}_0^T(\mathbf{A}\mathbf{x}_0 - \mathbf{b})}{\mathbf{p}_0^T\mathbf{A}\mathbf{p}_0} = -\frac{\mathbf{p}_0^T\nabla f(\mathbf{x}_0)}{\mathbf{p}_0^T\mathbf{A}\mathbf{p}_0}, \quad (2)$$

where the definition of $\nabla f(\mathbf{x}_0)$ has been substituted. $\square$

Exercise 4.2:

Consider minimizing, using the method of conjugate gradient, the cost function $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{A} \mathbf{x} - \mathbf{b}^T \mathbf{x}$ with respect to the design variable $\mathbf{x}$, where $\mathbf{A}$ is a positive definite matrix.

The search direction for the $(k+1)$ th iteration is given by

$$\mathbf{p}_k = -\nabla f(\mathbf{x}_k) + \beta_k \mathbf{p}_{k-1}. \quad (3)$$

Show that given that $\mathbf{p}_{k-1}$ is a descent direction at $\mathbf{x}_{k-1}$, $\mathbf{p}_k$ is a descent direction at $\mathbf{x}_k$ if $\beta_k \geq 0$.

Proof:

We express $\mathbf{x}_k$ as $\mathbf{x}_k = \mathbf{x}_{k-1} + a_{k-1} \mathbf{p}_{k-1}$, where $a_{k-1} = -\frac{\mathbf{p}_{k-1}^T \nabla f(\mathbf{x}_{k-1})}{\mathbf{p}_{k-1}^T \mathbf{A} \mathbf{p}_{k-1}} > 0$ is the step size (see (4.20a) and (4.20b) in the course notes).

Since $-\nabla f(\mathbf{x}_k)$ is already a descent direction at $\mathbf{x}_k$, we only need to show that $\mathbf{p}_{k-1}$ is not an ascent direction at $\mathbf{x}_k$. For this purpose, we have

$$\begin{aligned} \mathbf{p}_{k-1}^T \nabla f(\mathbf{x}_k) &= \mathbf{p}_{k-1}^T (\mathbf{A} \mathbf{x}_k - \mathbf{b}) \\ &= \mathbf{p}_{k-1}^T (\mathbf{A}(\mathbf{x}_{k-1} + a_{k-1} \mathbf{p}_{k-1}) - \mathbf{b}) \\ &= \mathbf{p}_{k-1}^T (\mathbf{A} \mathbf{x}_{k-1} - \mathbf{b}) + a_{k-1} \mathbf{p}_{k-1}^T \mathbf{A} \mathbf{p}_{k-1} \\ &= \mathbf{p}_{k-1}^T \nabla f(\mathbf{x}_{k-1}) + a_{k-1} \mathbf{p}_{k-1}^T \mathbf{A} \mathbf{p}_{k-1} \\ &= 0, \end{aligned} \quad (4)$$

where $a_{k-1} = -\frac{\mathbf{p}_{k-1}^T \nabla f(\mathbf{x}_{k-1})}{\mathbf{p}_{k-1}^T \mathbf{A} \mathbf{p}_{k-1}} > 0$ has been used in reaching the last equality (also see (2) and (4.20a) in the course notes). This indicates that $\mathbf{p}_{k-1}$ is orthogonal to $\nabla f(\mathbf{x}_k)$, and it is not an ascent direction. This completes the proof. $\square$

Exercise 4.3:

Consider minimizing, using the Newton's method, the cost function $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{A}\mathbf{x} + \mathbf{b}^T \mathbf{x} + c$ with respect to the design variable $\mathbf{x}$, where $\mathbf{A}$ is a positive definite matrix.

Suppose the initial solution guess is $\mathbf{x}_0$. Show that with Newton's method, we can converge to the optimal solution in one iteration.

Proof:

The optimal solution is $\mathbf{x}^* = -\mathbf{A}^{-1}\mathbf{b}$. This can be verified by taking the partial derivative of $f(\mathbf{x})$ with respect to $\mathbf{x}$ and setting the result to zero.

With Newton's method, we need to calculate the Hessian at $\mathbf{x}_0$, $\mathbf{H}(\mathbf{x}_0)$, which is equal to $\mathbf{A}$.

The next solution $\mathbf{x}_1$ can thus be obtained using

$$\begin{aligned}
 \mathbf{x}_1 &= \mathbf{x}_0 - \mathbf{H}(\mathbf{x}_0)^{-1} \nabla f(\mathbf{x}_0) \\
 &= \mathbf{x}_0 - \mathbf{A}^{-1}(\mathbf{A}\mathbf{x}_0 + \mathbf{b}) \\
 &= \mathbf{x}_0 - \mathbf{x}_0 - \mathbf{A}^{-1}\mathbf{b} \\
 &= -\mathbf{A}^{-1}\mathbf{b} \\
 &= \mathbf{x}^*,
 \end{aligned} \tag{5}$$

where $f(\mathbf{x}_0) = \mathbf{A}\mathbf{x}_0 + \mathbf{b}$ has been substituted to reach the third equality. This completes the proof. $\square$

Exercise 4.4:

Consider minimizing, using the Levenberg-Marquardt (LM) algorithm, the nonlinear least squares (NLS) cost function $f(\mathbf{x}) = \frac{1}{2}(\mathbf{b} - \mathbf{g}(\mathbf{x}))^T(\mathbf{b} - \mathbf{g}(\mathbf{x}))$ with respect to the design variable $\mathbf{x}$. Here, $\mathbf{b}$ is the measurement vector, and $\mathbf{g}(\cdot)$ is the nonlinear vector measurement function.

In iteration k , the LM algorithm first approximates $\mathbf{g}(\mathbf{x})$ using the first-order Taylor-series expansion as $\mathbf{g}(\mathbf{x}) \approx \mathbf{g}(\mathbf{x}_{k-1}) + \mathbf{J}(\mathbf{x}_{k-1})(\mathbf{x} - \mathbf{x}_{k-1})$, where $\mathbf{x}_{k-1}$ is the solution from the previous iteration and $\mathbf{J}(\mathbf{x}_{k-1})$ is the Jacobian matrix.

Putting the above approximation back to the NLS cost function and minimizing it yields the solution $\mathbf{x}_k$ in the current iteration, iteration k . Find $\mathbf{x}_k$.

Solution:

The NLS cost function becomes, after applying $\mathbf{g}(\mathbf{x}) \approx \mathbf{g}(\mathbf{x}_{k-1}) + \mathbf{J}(\mathbf{x}_{k-1})(\mathbf{x} - \mathbf{x}_{k-1})$,

$$\begin{aligned} f(\mathbf{x}) &\approx \frac{1}{2}(\mathbf{b} - \mathbf{g}(\mathbf{x}_{k-1}) - \mathbf{J}(\mathbf{x}_{k-1})(\mathbf{x} - \mathbf{x}_{k-1}))^T(\mathbf{b} - \mathbf{g}(\mathbf{x}_{k-1}) - \mathbf{J}(\mathbf{x}_{k-1})(\mathbf{x} - \mathbf{x}_{k-1})) \\ &\propto \frac{1}{2}(\mathbf{x} - \mathbf{x}_{k-1})^T \mathbf{J}(\mathbf{x}_{k-1})^T \mathbf{J}(\mathbf{x}_{k-1})(\mathbf{x} - \mathbf{x}_{k-1}) - (\mathbf{x} - \mathbf{x}_{k-1})^T \mathbf{J}(\mathbf{x}_{k-1})^T (\mathbf{b} - \mathbf{g}(\mathbf{x}_{k-1})), \end{aligned} \quad (6)$$

where the terms not depending on the unknown $\mathbf{x}$ have been ignored.

Because $\mathbf{x}_{k-1}$ is known, minimizing $f(\mathbf{x})$ in (6) with respect to $\mathbf{x}$ is equivalent to minimizing it with respect to $(\mathbf{x} - \mathbf{x}_{k-1})$.

As a result, the minimizer can be shown to satisfy

$$\mathbf{J}(\mathbf{x}_{k-1})^T \mathbf{J}(\mathbf{x}_{k-1})(\mathbf{x} - \mathbf{x}_{k-1}) - \mathbf{J}(\mathbf{x}_{k-1})^T (\mathbf{b} - \mathbf{g}(\mathbf{x}_{k-1})) = \mathbf{0}. \quad (7)$$

This is obtained by taking partial derivative of $f(\mathbf{x})$ in (6) with respect to $(\mathbf{x} - \mathbf{x}_{k-1})$ and setting the result to zero. Solving (7) for $\mathbf{x}$ yields the solution in the k th iteration, which is

$$\mathbf{x}_k = \mathbf{x}_{k-1} + (\mathbf{J}(\mathbf{x}_{k-1})^T \mathbf{J}(\mathbf{x}_{k-1}))^{-1} \mathbf{J}(\mathbf{x}_{k-1})^T (\mathbf{b} - \mathbf{g}(\mathbf{x}_{k-1})). \quad (8)$$

□

Exercise 4.5:

Consider minimizing the nonlinear least squares (NLS) cost function $f(\mathbf{x}) = \frac{1}{2}(\mathbf{b} - \mathbf{g}(\mathbf{x}))^T(\mathbf{b} - \mathbf{g}(\mathbf{x}))$ with respect to the design variable $\mathbf{x}$. Here, $\mathbf{b}$ is the measurement vector, and $\mathbf{g}(\cdot)$ is the nonlinear vector measurement function.

Derive the gradient of the cost function $f(\mathbf{x})$.

Solution:

We re-write the cost function as

$$f(\mathbf{x}) \propto \frac{1}{2}\mathbf{g}(\mathbf{x})^T\mathbf{g}(\mathbf{x}) - \mathbf{b}^T\mathbf{g}(\mathbf{x}), \quad (9)$$

where the terms independent of the design variable vector $\mathbf{x}$ are all ignored.

Recall that the Jacobian is defined as

$$\mathbf{J}(\mathbf{x}) = \frac{\partial \mathbf{g}(\mathbf{x})}{\partial \mathbf{x}} = \begin{bmatrix} \nabla g_1(\mathbf{x})^T \\ \nabla g_2(\mathbf{x})^T \\ \vdots \\ \nabla g_d(\mathbf{x})^T \end{bmatrix}, \quad (10)$$

with $\mathbf{g}(\mathbf{x}) = [g_1(\mathbf{x}), g_2(\mathbf{x}), \dots, g_d(\mathbf{x})]^T$.

As a result, we have

$$\frac{\partial \mathbf{g}(\mathbf{x})^T \mathbf{g}(\mathbf{x})}{\partial \mathbf{x}} = \frac{\partial \sum_{i=1}^d g_i(\mathbf{x})^2}{\partial \mathbf{x}} = 2 \sum_{i=1}^d g_i(\mathbf{x}) \nabla g_i(\mathbf{x}) = 2\mathbf{J}(\mathbf{x})^T \mathbf{g}(\mathbf{x}), \quad (11)$$

and

$$\frac{\partial \mathbf{b}^T \mathbf{g}(\mathbf{x})}{\partial \mathbf{x}} = \frac{\partial \sum_{i=1}^d b_i g_i(\mathbf{x})}{\partial \mathbf{x}} = \sum_{i=1}^d b_i \nabla g_i(\mathbf{x}) = \mathbf{J}(\mathbf{x})^T \mathbf{b}, \quad (12)$$

where $\mathbf{b} = [b_1, b_2, \dots, b_d]^T$ has been used.

Therefore, we arrive at

$$\nabla f(\mathbf{x}) = -\mathbf{J}(\mathbf{x})^T(\mathbf{b} - \mathbf{g}(\mathbf{x})). \quad (13)$$

□

Exercise 4.6:

In the quasi-Newton method, the cost function $f(\mathbf{x})$ is approximated in the k th iteration as

$$f(\mathbf{x}) \approx f(\mathbf{x}_k) + \nabla f(\mathbf{x}_k)^T (\mathbf{x} - \mathbf{x}_k) + \frac{1}{2} (\mathbf{x} - \mathbf{x}_k)^T \tilde{\mathbf{H}}(\mathbf{x}) (\mathbf{x} - \mathbf{x}_k), \quad (14)$$

where $\tilde{\mathbf{H}}(\mathbf{x})$ is the approximated Hessian of $f(\mathbf{x})$ at $\mathbf{x}_k$.

Find the gradient of the approximated cost function in (14) at $\mathbf{x}_{k-1}$.

Solution:

The gradient of the approximated cost function in (14) is

$$\nabla f(\mathbf{x}) = \nabla f(\mathbf{x}_k) + \tilde{\mathbf{H}}(\mathbf{x}) (\mathbf{x} - \mathbf{x}_k). \quad (15)$$

Setting $\mathbf{x} = \mathbf{x}_{k-1}$ in (15) yields the desired result

$$\nabla f(\mathbf{x}_{k-1}) = \nabla f(\mathbf{x}_k) + \tilde{\mathbf{H}}(\mathbf{x}) (\mathbf{x}_{k-1} - \mathbf{x}_k) = \nabla f(\mathbf{x}_k) - \tilde{\mathbf{H}}(\mathbf{x}) (\mathbf{x}_k - \mathbf{x}_{k-1}). \quad (16)$$

□

Exercise 4.7:

In the quasi-Newton method, the cost function $f(\mathbf{x})$ is approximated in the k th iteration as

$$f(\mathbf{x}) \approx f(\mathbf{x}_k) + \nabla f(\mathbf{x}_k)^T (\mathbf{x} - \mathbf{x}_k) + \frac{1}{2} (\mathbf{x} - \mathbf{x}_k)^T \tilde{\mathbf{H}}(\mathbf{x}) (\mathbf{x} - \mathbf{x}_k), \quad (17)$$

where $\tilde{\mathbf{H}}(\mathbf{x})$ is the approximated Hessian of $f(\mathbf{x})$ at $\mathbf{x}_k$.

Minimizing the above cost function yields the solution for the next iteration, which is

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k \tilde{\mathbf{H}}(\mathbf{x}_k)^{-1} \nabla f(\mathbf{x}_k), \quad (18)$$

where α_k is the step size.

Show that the search direction in (18), which is $\mathbf{p}_k = -\tilde{\mathbf{H}}(\mathbf{x}_k)^{-1} \nabla f(\mathbf{x}_k)$ is a descent direction.

Proof:

With the quasi-Newton method, the approximated Hessian $\tilde{\mathbf{H}}(\mathbf{x})$ is always positive definite. As a result, we have

$$\mathbf{p}_k^T \nabla f(\mathbf{x}_k) = -\nabla f(\mathbf{x}_k)^T \tilde{\mathbf{H}}(\mathbf{x}_k)^{-1} \nabla f(\mathbf{x}_k) < 0. \quad (19)$$

The last inequality comes from the positive definiteness of $\tilde{\mathbf{H}}(\mathbf{x})$, which indicates that the intersection angle between $\mathbf{p}_k$ and $\nabla f(\mathbf{x}_k)$ is larger than 90° , and $\mathbf{p}_k$ is a descent direction.

□

Exercise 4.8:

The derivative of a scalar function $f(x)$ can be numerically approximated using

$$\frac{df(x)}{dx} = \frac{f(x+h) - f(x-h)}{2h}, \quad (20)$$

which is an average of the forward difference and backward difference.

Show that the estimation error of (20) is $\mathcal{O}(h^2)$.

Proof:

We can the third-order Taylor series expansions

$$f(x+h) = f(x) + \frac{df(x)}{dx}h + \frac{1}{2!} \frac{d^2f(x)}{dx^2}h^2 + \frac{1}{3!} \frac{d^3f(x)}{dx^3}h^3 + \mathcal{O}(h^4), \quad (21)$$

and

$$f(x-h) = f(x) - \frac{df(x)}{dx}h + \frac{1}{2!} \frac{d^2f(x)}{dx^2}h^2 - \frac{1}{3!} \frac{d^3f(x)}{dx^3}h^3 + \mathcal{O}(h^4). \quad (22)$$

□

Therefore, we have

$$f(x+h) - f(x-h) = 2\frac{df(x)}{dx}h + \frac{2}{3!} \frac{d^3f(x)}{dx^3}h^3 + \mathcal{O}(h^4), \quad (23)$$

and

$$\frac{f(x+h) - f(x-h)}{2h} = \frac{df(x)}{dx} + \frac{1}{3!} \frac{d^3f(x)}{dx^3}h^2 + \mathcal{O}(h^3). \quad (24)$$

The estimation error, which is mainly $\frac{1}{3!} \frac{d^3f(x)}{dx^3}h^2$, is thus $\mathcal{O}(h^2)$.